

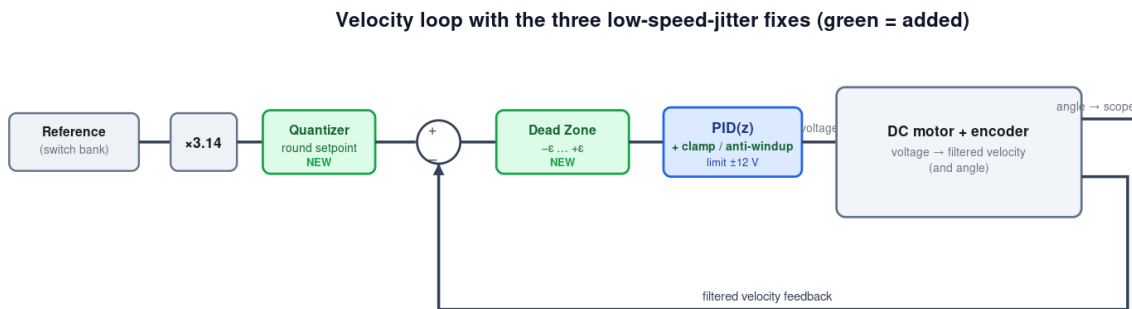
MSE 312 — Low-speed jitter fix for the velocity PID loop

Why the controller hunts near zero speed, and the three Simulink edits that stop it.

Why it happens

The velocity is estimated by differencing quantized encoder counts. At speed you get many counts per sample and a smooth estimate; near zero you get one or zero counts per sample, so the reading flickers between discrete values. The slider/ $\times 3.14$ setpoint is a long decimal that the quantized feedback can never exactly equal, so the integrator keeps chasing an unresolvable error — a limit cycle you see as jitter. It is worst from ~ 30 down to 0 and fades at higher speed because the velocity signal-to-noise recovers and friction/stiction no longer dominate.

The three edits (green blocks)



tolerance band ignores sub-quantization error · stops integrator windup · makes the target a reachable value
(the right-side velocity→angle integrator is a scope monitor only; not in the loop)

#	Edit	Block (library)	Where	Setting / why
1	Tolerance / deadband	Dead Zone (Discontinuities)	on the wire from Sum output into PID(z)	Start= $-\epsilon$, End= $+\epsilon$. Inside $\pm\epsilon$ the PID sees 0 error and stops hunting. For 'zero inside, full error outside' use Abs → Compare To Constant ($>\epsilon$) → Switch.
2	Anti-windup	PID(z) block options	inside the existing PID(z) block	Output Saturation tab: tick 'Limit output' (± 12 V), Anti-windup = clamping. Also try lowering I gain.
3	Reachable setpoint	Quantizer (Discontinuities) or Rounding Function	right after the $\times 3.14$ gain, before the Sum	Quantization interval = your velocity resolution. Rounds the target to a value the feedback can actually reach → true zero error.

Picking the tolerance ϵ

Hold the motor at a constant speed and look at the ripple on the **filtered velocity** signal. Set ϵ just above that ripple band. Start small and raise it until the low-speed jitter disappears. Trade-off: with the deadband the motor settles anywhere within $\pm\epsilon$ of the target rather than exactly on it — that is the price for killing the hunting, and for velocity hold it is the right trade.

Notes

- Best practice is to gate only the INTEGRATOR with the deadband (keep P always active), so small corrections still happen while the integrator stops accumulating unresolvable error. The PID(z) clamp in edit 2 largely covers this.
- The right-side velocity→angle Discrete-Time Integrator is a scope monitor, not part of the loop — it does not cause the jitter. But note it will DRIFT if the velocity has any DC bias, so do not trust it for absolute position over long runs.
- If the band edge still chatters, replace the Dead Zone with a Relay block to add hysteresis.
- A cleaner long-term fix is a better low-speed velocity estimate (time-between-edges / 1-over-dt instead of fixed-dt differencing), but the three edits above solve the immediate jitter.